

Research interests

My research is in Geometric Analysis and Nonlinear Partial Differential Equations. Most of it concerns fractional-order problems on compact Riemannian manifolds. For my doctoral work I study fractional Yamabe-type equations under isoparametric symmetry, using spectral theory for the fractional Laplacian $(-\Delta_g)^s$ together with variational tools such as the Symmetric Mountain Pass Theorem and Krasnoselskii genus theory. Regularity is handled through the Caffarelli–Silvestre extension. I am also interested in how the geometric structure of isoparametric foliations produces dimensional reduction and sharper Sobolev embeddings, and in bifurcation for the associated resolvent operators. Earlier I worked on the theory of viscosity solutions, including uniqueness and stability of saddle-shaped solutions to the Allen–Cahn equation.

Education

Aug 2023 - Present	Ph.D. in Mathematics Centro de Investigación en Matemáticas - CIMAT Title: TBA Advised by: Héctor A. Chang-Lara and Jimmy Petean
Mar 2022 - Jul 2023	M.Sc. in Mathematics Pontifícia Universidade Católica do Rio de Janeiro - PUC-Rio Title: Uniqueness of L^p -strong solutions Advised by: Boyan Sirakov and Pammella Queiroz
Jun 2015 - Jul 2019	B.Sc. in Mathematics Universidade Federal Fluminense - UFF

Fellowships

Aug 2023 - TBA	PhD Fellowship Consejo Nacional de Humanidades, Ciencias y Tecnologías - CONACYT
Mar 2022 - Jul 2023	Master Fellowship Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - CAPES Certificate

Aug 2017 - Aug 2018	Cientific Initiation Fellowship Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro - FAPERJ
Aug 2015 - Aug 2016	Cientific Initiation Fellowship Instituto de Matemática Pura e Aplicada - IMPA Certificate

Advanced Schools

Jan 2023 - Mar 2023	Summer course in Differential Topology Instituto de Matemática Pura e Aplicada - IMPA, Brasil Certificate
Jan 2022 - Feb 2022	Summer Program in Vectorial Analysis (Virtual) Universidade Federal de Pernambuco - UFPE, Brasil Certificate
Jan 2015 - Feb 2015	Summer course in Linear Algebra Instituto de Matemática Pura e Aplicada - IMPA, Brasil Certificate

Participations in Events

Nov 2024	Potential Theory Workshop: Intersections in Harmonic Analysis, Partial Differential Equations and Probability Centro de Investigación en Matemáticas - CIMAT, México Certificate
2017	31st Brazilian Mathematics Colloquium Instituto de Matemática Pura e Aplicada - IMPA, Brasil Certificate
2016	VIII UFF Math Week Universidade Federal Fluminense - UFF, Brasil

2015	2nd Workshop on Combinatorics, Number Theory and Dynamical Systems Instituto de Matemática Pura e Aplicada - IMPA, Brasil Certificate
2015	30th Brazilian Mathematics Colloquium Instituto de Matemática Pura e Aplicada - IMPA, Brasil Certificate
Nov 2014	Mathematics Biennial Universidade Federal Fluminense - UFF, Brasil
Nov 2014	Mathematics Week Universidade Federal Fluminense - UFF, Brasil